# Comparing Longstaff–Schwartz and Reinforcement Learning for American Option Pricing

A Numerical Comparison of LSMC and LSPI

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**Abstract**

This paper compares Longstaff–Schwartz Monte Carlo (LSMC) and Least-Squares Policy Iteration (LSPI), a reinforcement-learning method, for pricing an American put option. Both methods are tested under the same geometric Brownian motion model and compared with a Cox–Ross–Rubinstein binomial-tree benchmark. The experiment considers seven Monte Carlo sample sizes between 500 and 100,000 simulated stock-price paths, with each sample size repeated 20 times. The results show that LSMC becomes increasingly accurate as the number of paths rises, reaching a mean absolute percentage error of approximately 0.42% at 100,000 paths. LSPI also improves and becomes more stable as additional paths are provided, but its average pricing error begins to level off, remaining at approximately 4.54% at 100,000 paths. Examination of the stopping policy shows that LSPI learns a lower exercise boundary than the binomial benchmark, causing the option to be held in some states where the benchmark would exercise. Under the model and specification considered, LSMC therefore provides the more accurate estimate of the American put price.

## 1 Introduction

American options differ from European options because they can be exercised before maturity. For an American put, this means that at each possible exercise date the holder must choose between taking the current payoff or continuing to hold the option in the hope of receiving a larger payoff later.

The value of the contract therefore depends not only on future stock prices, but also on when exercise takes place. Finding the best time to exercise is known as an *optimal stopping problem*.

A widely used simulation-based approach to this problem is Longstaff–Schwartz Monte Carlo (LSMC), introduced by Longstaff and Schwartz [2]. Rather than constructing every possible future stock-price movement, LSMC generates many possible price histories and uses regression to estimate whether continuing to hold the option is more valuable than exercising it.

The same problem can also be viewed from a reinforcement-learning perspective. At each exercise date, the model observes the current situation and chooses between two actions: exercise or continue. Least-Squares Policy Iteration (LSPI), introduced by Lagoudakis and Parr [4], is a reinforcement-learning algorithm that can be used to learn such a decision rule.

This paper compares these two approaches under the same numerical setting. The main question is whether LSMC or LSPI produces an American put price closer to a binomial-tree benchmark as the number of simulated stock-price paths increases.

The comparison focuses not only on the final pricing error, but also on how the two methods respond to additional simulation data and whether differences in their learned exercise decisions can help explain their pricing performance.

## 2 Related Literature

The early-exercise feature of American options makes their numerical valuation more difficult than that of otherwise similar European options. A number of methods have therefore been developed to approximate both the value of the contract and the exercise strategy.

The Cox–Ross–Rubinstein (CRR) model [1] provides one of the standard approaches. The model represents future stock-price movements using a tree in which the price can move either upwards or downwards at each step. Starting from maturity and working backwards through the tree makes it possible to compare the value of immediate exercise with the value of continuing to hold the option at each point. For simple low-dimensional problems, a sufficiently fine tree provides a useful numerical benchmark.

Longstaff and Schwartz [2] introduced a simulation-based alternative that is particularly useful when lattice-based methods become difficult to apply. Their method generates possible future paths of the underlying asset and estimates the expected value of continuing to hold the option using least-squares regression. Exercise occurs when the immediate payoff exceeds this estimated continuation value. Their results show that the approach can provide accurate estimates for a range of American-style securities.

Related regression-based approaches were also studied by Tsitsiklis and Van Roy [3], who developed approximate dynamic-programming methods for pricing American-style options. Their work also uses regression to approximate values that are difficult to calculate directly, but approaches the stopping problem from a value-function perspective.

Egloff [6] further examined Monte Carlo methods for optimal stopping and their connection with statistical learning. This provides a broader theoretical link between simulation-based exercise decisions and learning methods of the type considered in the present comparison.

Reinforcement learning provides another way to view the same problem. Instead of directly estimating only the value of continuing, the problem can be treated as a sequence of decisions in which an algorithm learns the value associated with different actions. Lagoudakis and Parr [4] developed LSPI as a general reinforcement-learning algorithm that combines policy iteration with least-squares temporal-difference learning.

More recent research has extended learning-based optimal stopping methods using flexible machine-learning models. Becker, Cheridito and Jentzen [7], for example, use deep neural networks to learn stopping decisions in high-dimensional optimal stopping problems.

LSMC and LSPI therefore share some mathematical ideas, particularly the use of least-squares approximation, but apply them differently. LSMC is specifically structured around the decision between immediate exercise and continuation. LSPI is a more general decision-making algorithm that attempts to learn the value associated with each available action.

Glasserman [5] provides a wider treatment of Monte Carlo methods in financial engineering and discusses the additional difficulties created by early-exercise derivatives.

The purpose of the present experiment is not to propose a new pricing algorithm. Instead, it provides a controlled numerical comparison of LSMC and LSPI when both methods receive the same simulated information. In particular, the experiment examines whether increasing the amount of simulation data affects the two methods in the same way and whether differences in their exercise policies can help explain differences in pricing accuracy.

# 3 Model Setup

The experiment considers an at-the-money American put option. A put option gives its holder the right to sell the underlying stock at a fixed strike price $K$. Its immediate exercise payoff at time $t$ is $(K - S_t)^+$, where $S_t$ is the current stock price and $(x)^+ = \max(x, 0)$.

The parameters used throughout the experiment are shown in Table 1.

Table 1: Model parameters.

| Parameter | Symbol | Value |
|---|---|---|
| Initial stock price | $S_0$ | 100 |
| Strike price | $K$ | 100 |
| Risk-free rate | $r$ | 0.05 |
| Dividend yield | $q$ | 0 |
| Volatility | $\sigma$ | 0.20 |
| Maturity | $T$ | 1 year |
| Exercise intervals | $M$ | 50 |

The option begins at-the-money because its initial stock price and strike price are both 100.

There are 50 equally spaced exercise intervals, corresponding to a time step of 0.02 years. Since exercise is restricted to these dates rather than being continuously available, the numerical contract is technically Bermudan rather than fully American.

This restriction is applied consistently to LSMC, LSPI and the binomial benchmark, so all three methods value the same contract.

# 4 Stock Price Simulation

To compare the two pricing methods, possible future movements of the underlying stock price must first be generated. A single *simulated path* represents one possible evolution of the stock price from the present until the option's maturity. Generating many such paths produces a range of possible future price histories on which the exercise strategies can be trained and tested.

Under the risk-neutral measure, the stock price is assumed to follow geometric Brownian motion (GBM),

$$dS_t = (r - q)S_t\,dt + \sigma S_t\,dW_t. \tag{1}$$

Here, $r$ is the risk-free interest rate, $q$ is the dividend yield, $\sigma$ is the volatility of the stock and $W_t$ represents the random component of the price movement.

The simulated paths are generated directly from the exact GBM transition,

$$S_{t+\Delta t} = S_t \exp\left[\left(r - q - \frac{1}{2}\sigma^2\right)\Delta t + \sigma \Delta W_t\right], \qquad \Delta W_t \sim N(0, \Delta t). \tag{2}$$

Using the exact transition rather than an Euler approximation avoids introducing an additional source of discretisation error into the simulated stock-price process.

Basic checks are used to confirm that the simulation behaves as expected. All paths begin at $S_0 = 100$, remain positive, and the simulated mean terminal stock price is close to the theoretical risk-neutral expectation of approximately 105.13.

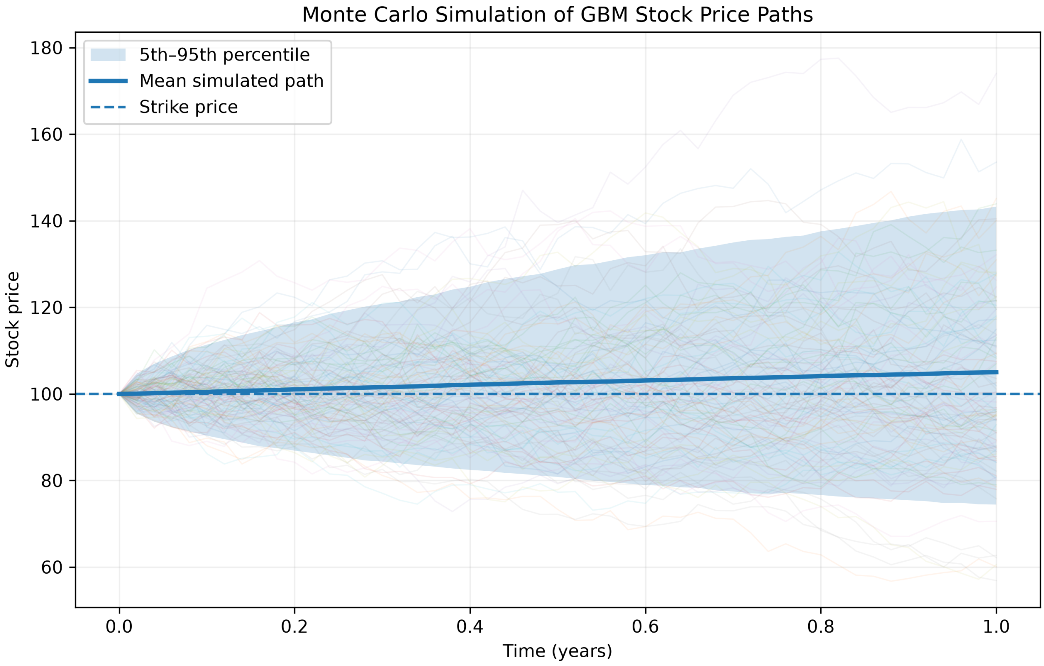


Figure 1: Monte Carlo simulation of possible GBM stock-price paths. The figure shows sample paths, the mean simulated path, the 5th–95th percentile range and the strike price.

Figure 1 illustrates the meaning of the simulated paths used throughout the experiment. Each thin trajectory represents one possible future evolution of the same stock under the assumed model.

# 5 Binomial Tree Benchmark

To judge the accuracy of LSMC and LSPI, a reference value is required. A Cox–Ross–Rubinstein binomial tree [1] is used for this purpose.

The binomial model approximates future stock-price movements using a tree in which the stock moves either upwards or downwards at each step. By working backwards from the option's maturity, the method can determine whether exercising or continuing is more valuable at each permitted exercise date.

For each tree step, the upward and downward movements are defined by $u = e^{\sigma\sqrt{\Delta t_{\text{tree}}}}$ and $d = 1/u$. The corresponding risk-neutral probability of an upward movement is $p = (e^{(r-q)\Delta t_{\text{tree}}} - d)/(u-d)$.

At each permitted exercise point, the value of the option is

$$V = \max\left\{(K-S)^+,\, e^{-r\Delta t_{\text{tree}}}\left[pV^u + (1-p)V^d\right]\right\}. \tag{3}$$

The first term represents the payoff from exercising immediately, while the second represents the discounted expected value of continuing to hold the option.

Because a numerical tree is itself an approximation, convergence of the benchmark is checked using several tree sizes.

Table 2: Convergence of the binomial benchmark.

| Tree steps | Option price |
|---|---|
| 500 | 6.0773 |
| 1,000 | 6.0781 |
| 2,500 | 6.0784 |
| 5,000 | 6.0785 |

The estimated price changes only slightly as the tree becomes larger. The 5,000-step result of $V_{\text{Binomial}} = 6.078525$ is therefore used as the fixed benchmark throughout the experiment.

Exercise in the tree is restricted to the same 50 dates used by LSMC and LSPI.

# 6 Longstaff–Schwartz Monte Carlo

LSMC works backwards through the simulated stock-price paths and estimates the *continuation value*: the expected value of keeping the option rather than exercising it immediately.

At each exercise date, this estimated continuation value is compared with the payoff available from immediate exercise.

The stock price is normalised as $x_t = S_t/K$, and the basis functions 1, $x_t$ and $x_t^2$ are used in the regression. The continuation value is approximated by

$$\widehat{C}_t(x) = \beta_{0,t} + \beta_{1,t}x + \beta_{2,t}x^2. \tag{4}$$

Only paths on which the put is currently in the money are included in the regression, since these are the states in which early exercise is relevant.

The decision rule is then intuitive. If the payoff available from exercising immediately is at least as large as the estimated continuation value, the option is exercised. Otherwise, it continues to be held.

Repeating this process backwards through all exercise dates produces an *exercise policy*: a rule describing when the option should be exercised under different stock-price conditions.

The fitted policy is then applied to a separate set of simulated paths. The estimated option price is the average discounted payoff produced by following that policy.

Using separate training and evaluation paths avoids measuring the performance of the stopping rule on exactly the same scenarios from which it was estimated.

# 7 Least-Squares Policy Iteration

LSPI represents the same exercise problem as a sequence of decisions. At each exercise date, the algorithm observes the current state of the option and chooses between two possible actions: exercise or continue.

The *state* contains the information available to the algorithm when making this decision. In the present experiment, it consists of the normalised stock price $x = S_t/K$ and normalised time $\tau = t/T$.

These variables are transformed into the feature vector

$$\psi(\tau, x) = \begin{bmatrix} 1 & x & x^2 & \tau & \tau x & (1-x)^+ \end{bmatrix}^\top. \tag{5}$$

Separate feature blocks are used for the exercise and continue actions, giving 12 action features in total.

LSPI estimates an *action value*, usually written as $Q(s, a)$, for each possible choice. This represents the estimated value of taking action $a$ when the option is in state $s$. The approximation used here is $Q(s, a) \approx \theta^\top \phi(s, a)$.

The policy therefore selects whichever of exercise or continue has the larger estimated action value.

Policy evaluation is carried out using Least-Squares Temporal Difference Q-learning (LSTDQ). In simplified form, the parameters are found by solving $A\theta = b$, where for continuing transitions

$$A = \sum_i \phi_i (\phi_i - \gamma \phi_i')^\top, \qquad b = \sum_i \phi_i r_i, \qquad \gamma = e^{-r\Delta t}. \tag{6}$$

The discount factor $\gamma$ accounts for the fact that a payoff received in the future is worth less than the same payoff received immediately.

A small regularisation term of $10^{-8}$ is included for numerical stability. LSPI repeatedly evaluates its current exercise policy and then updates the policy by choosing the action with the larger estimated value. This continues until the parameter vector changes by less than $10^{-6}$ or until 20 iterations have been completed.

As with LSMC, the final exercise policy is evaluated using an independent set of simulated stock-price paths.

# 8 Experimental Design

The central experiment examines how the accuracy of each method changes as more simulated stock-price paths become available. A larger number of paths gives each method more possible future market scenarios from which to estimate its exercise strategy.

For each repetition, a common set of GBM training paths is generated and used to fit both LSMC and LSPI. A second, independent set of paths is then generated to evaluate the two fitted policies. Both methods are therefore trained on the same scenarios and tested on the same new scenarios.

This use of common samples makes the comparison fairer, since differences in the results cannot simply be attributed to one method receiving more favourable random stock-price movements.

Seven training sample sizes are considered: 500, 1,000, 5,000, 10,000, 25,000, 50,000 and 100,000 paths.

Each sample size is repeated 20 times using different random seeds, giving 140 complete LSMC–LSPI comparisons. Repeating the experiment is important because Monte Carlo methods contain randomness, meaning that a single run could give an unusually good or poor result by chance.

For each repetition, pricing error is measured as the absolute difference between the estimated option value and the binomial benchmark. The principal measure reported for each sample size is the mean absolute error (MAE) across the 20 repetitions.

The error is also expressed as a percentage of the benchmark price. Error bars in the main results figure show approximate 95% confidence intervals for the mean percentage error, providing an indication of how much the results vary across repetitions.

# 9 Results

Table 3 reports the main numerical results. The "Price" columns show the average option-price estimate across the 20 repetitions, while MAE measures the average absolute distance from the binomial benchmark. The error standard deviation measures how much the pricing errors vary between repetitions.

Table 3: Mean pricing results across 20 repetitions.

| Paths | LSMC Price | LSMC MAE | LSMC Error SD | LSPI Price | LSPI MAE | LSPI Error SD |
|---|---|---|---|---|---|---|
| 500 | 5.8807 | 0.3924 | 0.2990 | 5.6873 | 0.5192 | 0.3968 |
| 1,000 | 5.9644 | 0.2722 | 0.1477 | 5.6690 | 0.4375 | 0.3251 |
| 5,000 | 6.0123 | 0.1117 | 0.0799 | 5.7689 | 0.3097 | 0.1359 |
| 10,000 | 6.0258 | 0.0669 | 0.0561 | 5.7801 | 0.2985 | 0.0938 |
| 25,000 | 6.0414 | 0.0460 | 0.0334 | 5.7924 | 0.2861 | 0.0558 |
| 50,000 | 6.0468 | 0.0345 | 0.0246 | 5.7956 | 0.2829 | 0.0376 |
| 100,000 | 6.0528 | 0.0257 | 0.0150 | 5.8024 | 0.2761 | 0.0186 |

The results show a clear improvement in LSMC as the number of paths increases. Its mean absolute error falls from 0.3924 with 500 paths to 0.0257 with 100,000 paths.

LSPI also improves, particularly at smaller sample sizes. Its mean absolute error falls from 0.5192 at 500 paths to 0.2985 at 10,000 paths. Beyond this point, however, the improvement becomes much smaller. Between 25,000 and 100,000 paths, mean absolute error decreases only from 0.2861 to 0.2761.

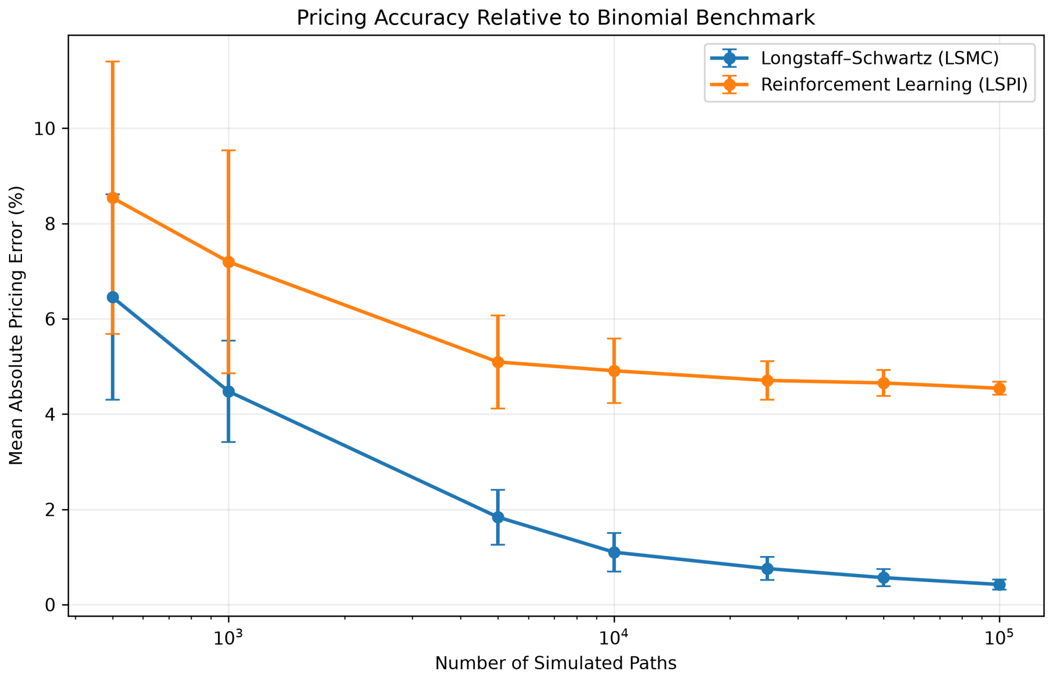


Figure 2: Mean absolute percentage pricing error relative to the binomial benchmark. Error bars show approximate 95% confidence intervals.

Figure 2 makes the difference in convergence more visible. At 500 paths, the mean percentage error is approximately 6.46% for LSMC and 8.54% for LSPI. At 100,000 paths, these errors have fallen to approximately 0.42% and 4.54%, respectively.

LSMC therefore continues moving towards the benchmark as the amount of simulation data increases.

LSPI also becomes considerably more stable. Its standard deviation falls and the confidence intervals become narrower as more paths are used. However, the average price does not continue moving towards the benchmark at the same rate. This suggests that the remaining difference is not simply the result of random variation between simulations.

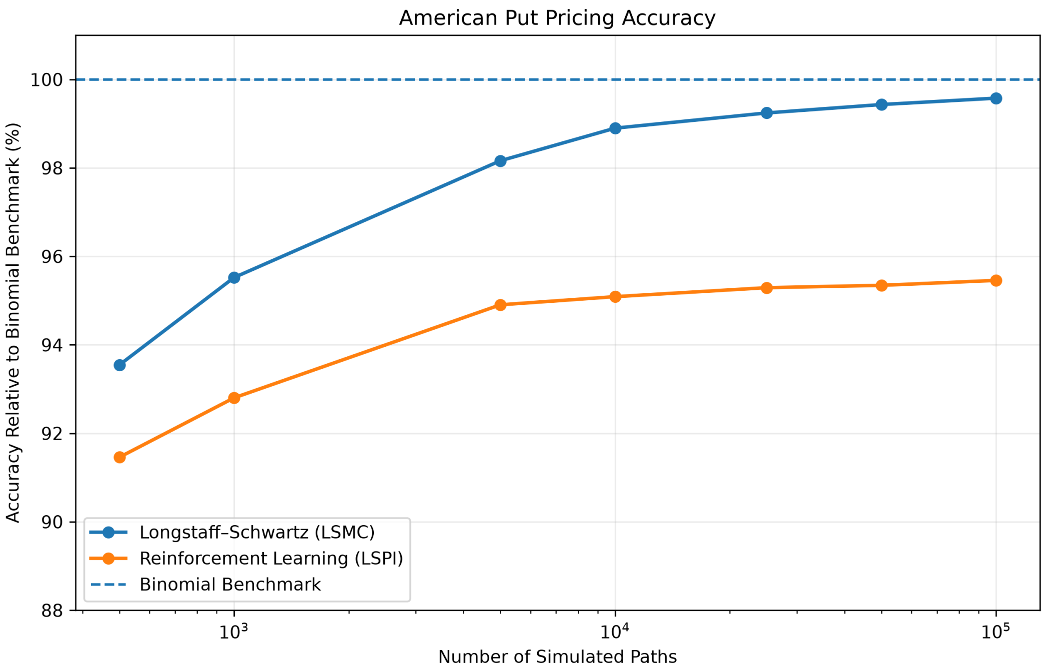


Figure 3: Pricing performance relative to the binomial benchmark, where 100% represents exact agreement.

The same result can be viewed in terms of benchmark agreement in Figure 3. At 100,000 paths, the LSMC estimate corresponds to approximately 99.58% agreement with the benchmark, compared with approximately 95.46% for LSPI.

## 10 Exercise-Policy Diagnostic

Pricing error shows that LSPI remains below the benchmark, but does not by itself explain why. The learned exercise policy provides an additional diagnostic.

An *exercise boundary* is the stock-price level separating states in which it is preferable to exercise the option from states in which it is preferable to continue holding it. For an American put, sufficiently low stock prices generally make immediate exercise more attractive.

Halfway through the option's life, at $t = 0.5$, the binomial tree gives an exercise boundary of approximately $S^*_{\text{Binomial}} \approx 84.87$. This means that the benchmark switches between exercise and continuation at a stock price of roughly 84.87 at that point in time.

On the stock-price grid used to inspect the LSPI policy, LSPI exercises at $S = 70$ but continues at $S = 75$. Its implied exercise boundary therefore lies somewhere between these two prices, substantially below the corresponding binomial boundary.

The LSPI policy consequently continues to hold the option for some stock prices at which the binomial benchmark would already exercise.

This matters because the value of an American option depends on the timing of exercise as well as the eventual payoff. Delaying exercise in states where immediate exercise is more valuable can reduce the value obtained from the early-exercise feature.

The exercise-boundary result therefore provides a possible explanation for the downward bias in the LSPI price estimates. It also supports the conclusion that the remaining difference is not caused entirely by random Monte Carlo noise: as the sample size increases, the LSPI estimates become increasingly stable while their average remains below the benchmark.

## 11 Discussion

The experiment shows that LSMC and LSPI respond differently as the amount of simulation data increases.

For LSMC, both the average pricing error and the variation between repetitions fall substantially. This behaviour is consistent with the role of least-squares regression in the original Longstaff–Schwartz method [2]. As more simulated observations become available, the regression has more information from which to estimate the value of continuing to hold the option.

The approach performs particularly well in the present setting because the problem contains only one stochastic state variable, the stock price, and the continuation value is relatively smooth. This is also consistent with the wider regression-based American option literature, where approximate value functions can perform effectively in relatively low-dimensional settings [3, 6].

LSPI also benefits from additional simulation data. Its pricing error falls substantially at smaller sample sizes and its estimates become much more stable. However, after approximately 10,000 to 25,000 paths, the improvement in average pricing error becomes much smaller.

This distinction is important. If the main problem were simply insufficient simulation data, increasing the number of paths would be expected to make the estimates both more stable and progressively closer to the benchmark. Instead, LSPI becomes more consistent while remaining systematically below it.

This result can be considered in relation to the purpose of LSPI in Lagoudakis and Parr [4]. LSPI was developed as a general method for learning policies in sequential decision problems rather than specifically as an American option pricing algorithm. Its performance therefore depends partly on whether the selected features are able to represent the values associated with the relevant states and actions.

In contrast, LSMC is designed specifically around the structure of the American option problem. It directly estimates the continuation value required for the exercise decision. This closer connection between the regression target and the underlying pricing problem may help explain its stronger performance in this relatively simple one-dimensional setting.

The exercise-policy diagnostic supports this interpretation. LSPI learns an exercise boundary below the binomial benchmark at $t = 0.5$, meaning that the policy delays exercise over part of the state space. This behaviour is consistent with the persistent downward pricing bias observed at larger sample sizes.

More flexible learning methods may be able to represent stopping policies more accurately. For example, Becker, Cheridito and Jentzen [7] demonstrate that deep-learning methods can be used to learn optimal stopping decisions in more complex and higher-dimensional settings.

The results should not be interpreted as evidence that reinforcement learning is generally inferior for American option pricing. The comparison concerns one underlying stock-price model, one option specification and one LSPI feature representation. Alternative features or more flexible reinforcement-learning algorithms could produce different results.

The more limited conclusion is that, under the conditions tested here, increasing the number of simulated paths is sufficient to make LSMC highly accurate, while additional paths alone do not remove the remaining error in the LSPI specification.

## 12 Limitations and Possible Extensions

Several limitations restrict how widely the results can be generalised.

First, the stock-price process follows geometric Brownian motion, which assumes constant volatility and lognormal returns. Real financial markets can display more complicated behaviour. A natural extension would be to repeat the experiment under a stochastic-volatility model such as the Heston model, in which volatility is itself allowed to change randomly over time.

Second, the experiment considers only one at-the-money put option. Testing different strike prices, maturities and volatility levels would show whether the same relative performance appears across a wider range of contracts.

Third, exercise is restricted to 50 dates rather than being continuously available. The numerical contract is therefore technically Bermudan, although the same restriction is imposed on every pricing method and does not affect the fairness of the comparison.

Finally, the LSPI results depend on the feature representation used to approximate its action values. Alternative basis functions or more flexible reinforcement-learning methods could be investigated to determine whether the learned exercise boundary moves closer to the benchmark. Recent work using neural networks for optimal stopping suggests one possible direction for such an extension [7].

These extensions would provide a broader test of the results but would also substantially increase the scope of the experiment.

## 13 Conclusion

This paper compared Longstaff–Schwartz Monte Carlo and Least-Squares Policy Iteration for pricing an American put option under geometric Brownian motion. Both methods were trained using common simulated stock-price paths, evaluated using common independent test paths, and compared with a 5,000-step binomial-tree benchmark.

The results show that LSMC produces a lower mean absolute pricing error at every sample size considered. At 100,000 paths, its mean absolute percentage error is approximately 0.42%, compared with 4.54% for LSPI.

The difference between the two methods becomes particularly clear as the amount of simulation data increases. LSMC continues to move towards the benchmark while the variation between repetitions also falls. LSPI improves strongly at smaller sample sizes and becomes considerably more stable, but its average pricing error begins to level off.

Examining the exercise strategy provides a possible explanation for this behaviour. At $t = 0.5$, the exercise boundary learned by LSPI lies below the corresponding binomial boundary. The LSPI policy therefore continues to hold the option in some states where the benchmark indicates that immediate exercise is preferable.

The experiment consequently identifies a difference not only in final pricing accuracy but also in how the two methods respond to additional simulation data. Under the model and specification considered, Longstaff–Schwartz provides the more accurate and increasingly reliable estimate of the American put price. The remaining LSPI error appears to reflect a limitation of the learned policy under the chosen specification rather than a lack of simulated observations alone.

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